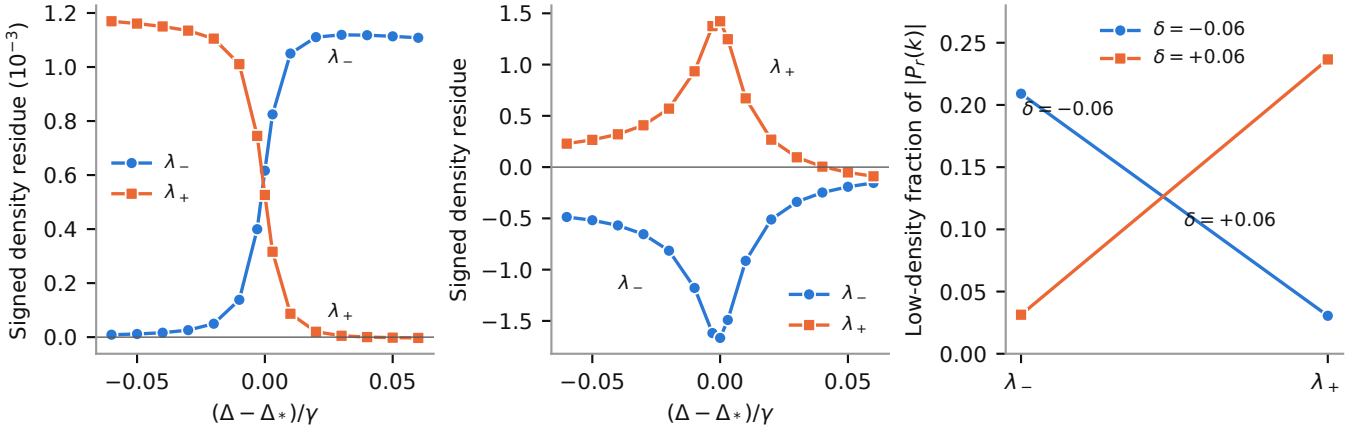
(a) High-branch perturbation
Signed density residue (10^{-3})
1.2
1.0
0.8
0.6
0.4
0.2
0.0
λ_-
λ_+
λ_-
λ_+
−0.05
0.00
0.05
$(\Delta - \Delta_*)/\gamma$
(b) Fold-state preparation
Signed density residue
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
λ_+
λ_-
λ_-
λ_+
−0.05
0.00
0.05
$(\Delta - \Delta_*)/\gamma$
(c) Profile character exchanges
Low-density fraction of $|P_r(k)|$
0.25
0.20
0.15
0.10
0.05
0.00
$\delta = -0.06$
$\delta = +0.06$
$\delta = -0.06$
$\delta = +0.06$
λ_-
λ_+